Workshop Data Platform
Vector Search Developer Guide

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1. Storing embeddings in a VECTOR column

Embeddings are stored in a column of type VECTOR. Declare the element type and the fixed length when you create the table, for example VECTOR(DOUBLE, 384) for the all-MiniLM-L6-v2 model. Every vector in that column must have exactly that number of elements.

To insert a vector from application code, pass it as a comma-separated string parameter and convert it with TO_VECTOR(?, DOUBLE, 384). Keep the original text, a document ID and a chunk ID in the same row so that search results can show where each passage came from.

Use the same embedding model for stored documents and for search questions. Vectors produced by different models live in different spaces and must never be compared with each other.

2. Choosing a similarity function

The platform supports two similarity functions in SQL: VECTOR_COSINE and VECTOR_DOT_PRODUCT. Cosine similarity compares only the direction of two vectors and returns a value between -1 and 1, where higher means more similar.

The dot product also depends on vector length. When every vector is normalized to length 1, the dot product and cosine similarity give the same ranking. Sort by the similarity score in descending order to get the closest matches first.

A similarity score is a ranking signal, not a probability. A score of 0.6 from one model cannot be compared with a score of 0.6 from another model, and the best match in a table can still be irrelevant to the question.

3. Speeding up vector search with an HNSW index

Without an index, a vector query compares the question vector with every stored vector. This exact search is always correct, but its cost grows with the size of the table.

An HNSW index is an approximate nearest neighbor index. It navigates a graph of neighboring vectors, so it compares far fewer vectors, but it can occasionally miss one of the true nearest neighbors. Create it with CREATE INDEX ... AS HNSW(Distance='Cosine') and make sure the distance matches the function used in the query.

The index is only considered for queries that use TOP, ORDER BY a vector similarity function, and DESC. Always check the query plan, and measure both latency and recall on your own data before relying on the index.

4. Connecting from Python with the DB-API driver

Python applications connect with the official InterSystems DB-API driver, installed from the package intersystems-irispython and imported as iris. Call iris.connect with the host name, port, namespace, user name and password.

Always pass values as query parameters with question-mark placeholders instead of building SQL strings. This avoids SQL injection and lets the database reuse the cached query plan.

Keep credentials out of source code. Read them from environment variables or a local configuration file that is excluded from version control.